

Multimodal Medical Assistant for Image-Text Clinical Triage

Abstract

This project implements a Multimodal Medical Assistant that reads five image-note pairs and writes five conversation JSON files (HPPCS[04]). The project brief lists Llama, BiomedCLIP or MedCLIP, BioBERT, OpenCV and LangChain because they do different jobs. OpenCV, PubMedCLIP and BioBERT turn pixels and the prescription into ranked findings. LangChain does not diagnose; it orchestrates: named prompt templates, a retriever over those findings, and LCEL chains that compose retrieve then generate. Llama 3.2 3B (text-only) is the sole generative LLM. On five synthetic cases, exact triage accuracy was 1.00 and mean keyword recall was 0.866. This is an educational prototype, not a diagnostic device.

1. Introduction

A multimodal assistant must both extract evidence from an image and a note and then talk about that evidence. Mixing those jobs into one undifferentiated script hides mistakes (for example dumping a histogram into a small LLM). The brief therefore names specialised encoders and an orchestrator (LangChain) as well as Llama.

2. Problem Statement

Given an image and a short note, produce a multi-turn conversation that summarises the case, correlates retrieved image-text findings with symptoms, assigns triage (emergency, urgent, soon or routine) and asks follow-up questions, with a non-clinical disclaimer.

3. Objectives

- Emit five conversation JSON files from five image-note pairs.
- Use Llama 3.2 as the generative LLM named in the brief.
- Use OpenCV, PubMedCLIP/MedCLIP and BioBERT as the multimodal front end.
- Use LangChain for prompt templates, cross-modal finding retrieval and chain composition.

4. Methodology

Why LangChain is in the brief. LangChain is not a medical model and does not raise diagnostic accuracy by itself. The brief lists it to integrate text and image pipelines and to orchestrate cross-modal queries and retrieval. That is software architecture: (1) PromptTemplate objects make the entity, reasoning and dialogue prompts explicit and reusable; (2) a retriever treats CLIP+BioBERT fused findings as documents and, given the patient note as query, returns only the top-k evidence to Llama; (3) LCEL chains encode the order retrieve then prompt then generate. Without that layer, a student either pastes the whole visual dictionary into the LLM (noise) or hard-codes prompts in many places (hard to mark). Llama still runs on local Ollama; a thin LangChain LLM wrapper keeps keep_alive and JSON repair.

Workflow. OpenCV describes the PNG. PubMedCLIP scores image prompts; BioBERT scores the same prompts against the note; the requested study locks modality. LangChain retrieves the top fused findings. Llama extracts entities, writes triage JSON from retrieved evidence, then writes four conversation turns. Note-based calibration stops over-calling emergency. BiomedCLIP was attempted; Hugging Face returned 401 on this machine, so PubMedCLIP (MedCLIP family) was used.

5. System Design / Implementation

Architecture is described in text. `main.py` is the entry point. `multimodal_encoder.py` hosts PubMedCLIP and BioBERT. `image_analyzer.py` runs OpenCV. `langchain_orchestrator.py` holds PromptTemplates, CrossModalFindingRetriever and LCEL chains. `ollama_client.py` is the local generate API. `medical_assistant.py` fuses stages and writes JSON. `dataset_builder.py` stores synthetic pairs.

6. Results and Analysis

Five conversations were produced. After modality lock, retrieved findings matched the intended family in every case. Exact triage accuracy was 5/5. Mean keyword recall was 0.866. Misses were wording gaps, not wrong-system errors. Metrics are educational keyword matches on synthetic data. LangChain's contribution is a smaller, labelled evidence pack for Llama, not a change in encoder weights.

7. Conclusion and Future Work

The assistant meets HPPCS[04] with Llama 3.2, the brief's encoders, and LangChain as orchestrator. Image understanding comes from OpenCV and CLIP, note alignment from BioBERT, generation from Llama, and retrieval/prompt wiring from LangChain. Limits: CLIP on synthetic images, Hub access to BiomedCLIP, and one generative LLM whereas the general PDF asks for two. Future work: BiomedCLIP on real de-identified studies.

9. References

[1] HAAI++ capstone instructions and HPPCS[04] brief. [2] A. Grattafiori et al., Llama 3, arXiv:2407.21783. [3] S. Eslami et al., PubMedCLIP. [4] J. Lee et al., BioBERT, 2020. [5] LangChain, <https://python.langchain.com> [6] Ollama, <https://ollama.com> [7] G. Bradski, OpenCV, 2000.

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